

# Quiz 2

## 601.471/671 NLP: Self-Supervised Models

Johns Hopkins University  
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Complete all questions.

Use additional paper if needed.

Time: 70 minutes.

Name of student: \_\_\_\_\_

## Q1. Concepts, Foundations and Big Picture

40 points

For each of the following questions, select the most appropriate answer. You may need to indicate multiple answers when you see “choose all that apply”.

1. (2 points) Which of the following best describes the purpose of subword tokenization in natural language processing?
- ☐ To split words into individual letters for easier processing.
  - ☐ To identify and separate common prefixes and suffixes in words.
  - ☐ To break down words into smaller meaningful units or subwords, which can help handle out-of-vocabulary words.
  - ☐ To count the frequency of each word in a text.

**Solution: 3**

2. (2 points) How does the Byte Pair Encoding (BPE) algorithm construct a vocabulary for a language model that is about to be trained?
- ☐ It constructs a vocabulary by randomly selecting words from the training corpus.
  - ☐ It constructs a vocabulary by selecting the most frequent words from the training corpus.
  - ☐ It constructs a vocabulary by starting with individual characters and iteratively merging the most frequent pair of symbols.
  - ☐ It constructs a vocabulary by starting with individual words and iteratively splitting the least frequent words into smaller units.

**Solution: 3**

3. (2 points) Which of the following best describes the bottleneck of recurrent neural networks (RNN) and how Attention provides a way to get around this bottleneck?
- ☐ RNNs struggle to learn from long sequences, where it gets challenging for the network to preserve remote information. Attention mitigates this by attending to the relevant parts of the input instead of relying on continuous hidden states.
  - ☐ RNNs are slow because of recursively computing the hidden states. Attention improves this by reducing the computable parameters.
  - ☐ RNNs suffer from the inability to learn from small datasets effectively. Attention mechanism addresses this by introducing additional learnable parameters to improve the model’s learning capability.
  - ☐ The bottleneck of RNNs is their lack of transparency in their internal operations. Attention solves the interpretability issue.

**Solution: 1**

4. (2 points) Why can’t we parallelize the computations of the RNN language model at training time?
- ☐ Because recurrent neural networks are too complex to be parallelized.
  - ☐ Because the computations of one-time step in a recurrent neural network depend on the computations of previous time steps, which prevents parallelization.
  - ☐ Because parallelization interferes with the backpropagation process during training.
  - ☐ Because neural networks require sequential data processing.

**Solution: 2**

5. (2 points) Which of the following about RNNs is incorrect?
- ☐ RNN suffers from the “vanishing gradient” problem, making them hard to train.
  - ☐ Unlike Transformers, the computation of RNN cannot be parallelized.
  - ☐ The computation cost of RNN is  $O(n)$  while Transformers is  $O(n^2)$  where  $n$  indicates sequence length. Therefore, RNN is more compute efficient than transformers when the input length is very long.
  - ☐ RNN cannot handle sequences of arbitrary length.

**Solution:** *The 4th one.*

6. (3 points) In Transformers, what will happen if you do not down-scale the attention scores before Softmax? (choose all that apply)
- ☐ The variance of the attention scores will be too high.
  - ☐ The resulting attention probabilities will all near zero, except one element that will be near 1. ☐
- This would make the activations too large, making training to be unstable.
- ☐ This would likely make the variance in the gradients too large, making the model converge slower.

**Solution:** *The first and the second.*

7. (2 points) Which of the following is incorrect about Attention?
- ☐ In encoder self-attention, the tokens have access to future tokens during computation
  - ☐ In decoder self-attention, future tokens are masked.
  - ☐ Attention is often followed by a normalization layer.
  - ☐ Cost of generating  $Q, K, V$  vectors grows quadratically with sequence length.

**Solution:** *4*

8. (2 points) Which is incorrect about Transformers?
- ☐ The query, key, and value in Transformers work similarly to a search operation where the query is the user request, the value is the search base, and the key is the output extracted.
  - ☐ The dimensions of the input and output of the attention mechanism in the Transformer model are *always* the same.
  - ☐ Normalization in the Transformer model is aimed to standardize the inputs and to stabilize the learning process.
  - ☐ The memory footprint of Transformers increases quadratically as a function of input length.

**Solution:** *1*

9. (2 points) Which of the following best describes training parallelization in a Transformer?
- ☐ Transformers utilize Self-Attention mechanisms that do not require the sequential processing of RNNs, thereby allowing for parallel processing of the input data.
  - ☐ Transformers parallelize the processing of different layers, which allows Self-Attention and Normalization to be computed simultaneously.
  - ☐ Transformers parallelize training by distributing computational tasks across multiple neural networks.
  - ☐ Transformers use alignment techniques to parallelize training.
  - ☐ Transformer parallelizes training by increasing the number of layers in the network.

**Solution:** *1*

10. (2 points) Which does not use an encoder layer?
- ☐ BERT
  - ☐ T5
  - ☐ GPT-3
  - ☐ RoBERTa

**Solution:** *3*

11. (2 points) Which of the following is incorrect about the BERT model?
- ☐ BERT produces contextualized embeddings: the embeddings of each word depend on its bidirectional context.
  - ☐ BERT is a Recurrent Neural Network.
  - ☐ In BERT, words can simultaneously incorporate information from its left and right context for learning a representation.

- Besides language modeling, BERT can also be used to perform classification tasks (e.g., Sentiment Classification).

**Solution:** *The 2nd one.*

12. (2 points) Which of the following is correct about the BERT model?
- The training objective of BERT is only masking words in sentences and letting the model predict them.
  - In BERT, the query, key, and value vectors remain constant throughout training.
  - Dramatically increasing the masking ratio of BERT *always* improves its performance as it utilizes the training data more efficiently.
  - Drastic reduction of the masking ratio of BERT will increase the training time.

**Solution:** *The 4th one.*

13. (3 points) Which are valid motivations for natural language prompting of LMs instead of fine-tuning them? (select all that apply)
- ☐ Prompting results in multiple copies of model parameters for each training task, which creates redundancy.
  - ☐ Prompting leads to consistent results.
  - ☐ Prompts are more interpretable.
  - ☐ Fine-tuned parameters are more interpretable.
  - ☐ Prompting requires backward computation of gradients.
  - ☐ Fine-tuning requires backward computation of gradients.

**Solution:** *3, 6*

14. (3 points) Which of the following are incorrect on In-Context Learning? (select all that apply)
- ☐ The model undergoes no weight updates during the process.
  - ☐ Models exhibit a bias towards more recently provided examples.
  - ☐ Models exhibit a bias towards the examples at the beginning of the prompt.
  - ☐ If all demos for a task were "positive," the model is less likely to produce a "positive" output for the same task compared to when the model is given demos that were "negative".
  - ☐ In-context learning is one of the methods of efficient parameter tuning.

**Solution:** *3,4,5*

15. (3 points) Which one(s) are correct about natural language prompts? (select all that apply)
- ☐ Models always interpret language prompts like humans do.
  - ☐ Models are sensitive to the ordering of the instances matter.
  - ☐ Models are sensitive to the phrasing of prompts.
  - ☐ Models are biased to generating popular outputs (words).
  - ☐ With scale, performance *always* increases.
  - ☐ The success of discrete (text) prompts is a progress toward interpretable human-AI communication.

**Solution:** *All correct except the 1st, 5th*

16. (3 points) Which one(s) are correct about continuous prompts? (select all that apply)
- ☐ Continuous prompts are used to guide the model's responses in a conversation.
  - ☐ Continuous prompts are as interpretable as discrete (text) prompts.
  - ☐ Continuous prompts are parameter-efficient compared to whole model tuning.
  - ☐ Continuous prompts are computationally efficient compared to whole model tuning.
  - ☐ Continuous prompts are a type of prompt that is used to generate a sequence of questions.
  - ☐ Continuous prompts are used to reset the context in a conversation with the model.

**Solution:** 1,3

17. (3 points) Which of the following statement(s) is correct regarding parameter efficient tuning? (select all that apply)
- ☐ Parameter-efficient tuning methods are more memory-efficient than head-tuning.
  - ☐ Parameter-efficient tuning methods are more memory-efficient than whole-model tuning.
  - ☐ Parameter-efficient tuning methods are more computationally-efficient than head-tuning.
  - ☐ Parameter-efficient tuning methods are more computationally efficient than whole-model tuning.

**Solution:** 2

## Q2. Short-answer questions

20 points

For each of the following questions, please briefly answer in 3-5 sentences.

1. (4 points) Give two reasons why we might prefer a character-based tokenization over a word-based tokenization when training a neural language model.

**Solution:** 1. *Handling rare words and misspelled words:* Character-based tokenization does not rely on a predefined vocabulary like word-based tokenization and can deal with rare and unknown words.  
2. *Language independent:* character-based tokenization is less dependent on a specific language than word-based tokenization, as it doesn't need to understand the morphological structure of the language. This makes it more suitable for multilingual applications.  
3. *Computational Efficiency:* Word-based tokenization can result in a large vocabulary size if the training data is huge, leading to more computational resources. On the other hand, character-based tokenization limits the vocabulary size to the different characters, which is much smaller even for large datasets. However, character-based models may require more time to train as they tend to have longer sequences.

**Your answer:**

2. (4 points) Give two reasons why we might prefer a word-based tokenization over a character-based tokenization when training a neural language model.

**Solution:** 1. *Semantic Understanding:* Word-based models usually better understand context and semantics since they work with whole words rather than individual characters. This allows them to predict larger units of text better and may result in more coherent and meaningful sentences in tasks such as text generation.  
2. *Shorter Sequences:* Compared to character-based models, word-based models deal with shorter sequences, which makes training faster and computation more efficient. Character-based models can require vast amounts of computational power and memory for huge and complex texts.

**Your answer:**

3. (4 points) A Student wants to train a language model on a noisy corpus and decides to train a language model and filter out sentences with low predicted probability (high perplexity) predicted by the trained model. The student noticed that there was a problem with this approach. What kind of sentences does

this approach favor to filter out? What would be the solution to this problem? (less than 4-5 sentences)  
Hint: think about sentence length.

**Solution:** *This approach tends to favor shorter sentences. The perplexity of a sequence generally increases with its length because the model has more opportunities to make errors on longer sequences than on shorter ones. Thus, a long sentence may have a high perplexity simply due to its length, even if it's perfectly grammatical and sensible. Conversely, shorter sentences, even if they consist of nonsensical or noisy phrases, might end up with lower perplexity. One possible solution to this problem would be to normalize the perplexity by the sentence length. This way, the perplexity measure is less biased toward shorter sentences and gives a more fair comparison across sentences of varying lengths.*

**Your answer:**

4. (4 points) You're given a language model that is currently set to use greedy decoding (selecting the word with the highest probability). When decoding from this model, why or why not might you consider using (a) nucleus (top-p) sampling, (b) temperature of 1, (c) temperate higher than 1?

**Solution:** *(a) Nucleus (top-p) sampling can be beneficial as it introduces randomness into the model's outputs while maintaining coherence. This can lead to more contextually relevant outputs, as the model is not just randomly picking words, but is considering a subset of words most likely to follow the current context. However, if the threshold is set too low, it might result in less diverse outputs. (b) A temperature of 1 can make the model's output more diverse. The model will sample words according to their probabilities, producing more varied and creative outputs. (c) A temperature higher than one makes the model's output even more random. While this can generate diverse outputs, it can also lead to less coherent or contextually relevant outputs.*

**Your answer:**

5. (4 points) OpenAI showed that GPT-3 can perform simple arithmetic (e.g., addition, multiplication) without any fine-tuning (i.e., in the zero/one-shot/few-shot setting). In Section 3.9.1 of their paper, they measure how many of the 3-digit addition problems in their test set also occur in GPT-3's training data by searching for all occurrences of the following expressions: "<NUM1> + <NUM2> =", and "<NUM1> plus <NUM2>". Using this method, they find that only 17 out of 2000 addition problems occur within the training data. Does this experiment convince you that the model has learned the underlying process of addition instead of just memorizing things from its training data? Why or why not?

**Solution:** The experiment does suggest that GPT-3 isn't merely regurgitating memorized this particular addition pattern from its training data. However, it doesn't definitively prove that it has learned the underlying process of addition. The model may have seen other patterns and examples of addition problems during training. More comprehensive experiments would be needed to convincingly demonstrate that the model has a robust understanding of the addition process, such as testing it on addition problems with larger numbers or with more digits or even on subtraction, multiplication, and division problems to see if it can apply the same concept of arithmetic.

**Your answer:**

### Q3. Transformer Language Models

20 points

Let's delve into the Transformer architecture.

- (8 points) Define the Self-Attention operation and compute its computational complexity when applied to a sequence  $n$  inputs  $\mathbf{x}_1, \dots, \mathbf{x}_n$  where  $\mathbf{x}_i \in \mathbb{R}^d$ . Use matrix notation when providing your definition. In your notation, show the dimensions of the intermediate variables and the output of the attention block.

**Solution:** Let the input tokens be  $\mathbf{X}$ , the key, query, and value are given by

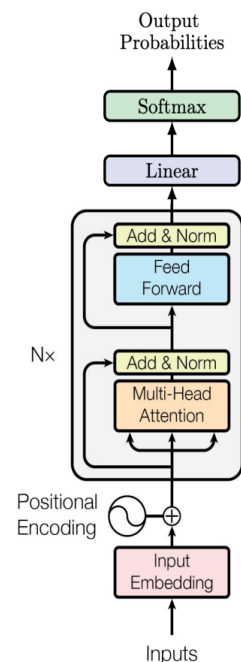
$$\mathbf{K} = \mathbf{X}\mathbf{W}^k, \mathbf{Q} = \mathbf{X}\mathbf{W}^q, \mathbf{V} = \mathbf{X}\mathbf{W}^v$$

Then the self-attention is given by

$$\mathbf{S} = \text{softmax} \left( \frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_q}} \right) \mathbf{V}$$

Regarding the dimensions, two things are important to show: (1) The dimensions are  $\mathbf{Q}\mathbf{K}^\top$  are  $n \times n$  where  $n$  is the sequence length. (2) The output dimension is the same as the output.

**Your answer:**



- (6 points) Describe how an input string is turned into a format that is readable by Transformer. In your description, discuss the role of tokenization, one-hot vectors, and embedding matrix  $E \in \mathbb{R}^{d \times V}$  where  $d$  is embedding size and  $V$  is the number of items in your vocabulary.



**Solution:** (1) The first stage is tokenization, in which a string of characters is broken into pieces or "tokens". The key is that each token gets turned into a unique integer. (2) After tokenization, the integers are translated into one-hot vectors. In this step, we convert each token index into a binary vector of size  $V$  where  $V$  is the total number of unique tokens (i.e., the vocabulary size). The vector is filled with 0s except at the index corresponding to the token, where we put a 1. For example, if the token index for "apple" is 453 and our vocabulary size  $V$  is 10,000, then the one-hot vector for "apple" is a vector of length 10,000 with 0s everywhere except at position 453. (3) Lastly, the one-hot vectors are multiplied by  $E$ , the embedding matrix. The multiplication operation can be thought of as a lookup that maps each one-hot vector to its embedded representation.

**Your answer:**

3. (6 points) Explain why "positional encodings" are needed and how they are added. Base your response to the content we saw in the class when discussing Transformers.

**Solution:** The purpose of the Transformer model is to process sequential data like language. Transformer models treat input tokens independently and don't have an inherent notion of the position of the words in a sentence. This can be problematic because the order of words in a sentence is meaningful – "cat eats fish" has different meaning from "fish eats cat". To solve this problem, the concept of "positional encodings" is introduced in the Transformer model. Positional encodings are added to the embedding of the tokens to inject information about the relative or absolute position of the tokens in the sequence. The idea is to generate a unique representation for each time step to maintain the relative information between different positions in the sequence.

**Your answer:**

## Q4. Adapting Language Models + Alignment

20 points

1. (3 points) You are given a pre-trained language model. You prompt this model:

Paragraph: "Among five hundred patients visited in Johns Hopkins Hospital .....". Find the number in the following paragraph and turn it into digits.

Which of the following outputs is likelier to be the output of this model and why?

**Output 1:**

The number is the number of patients in the hospital ...

**Output 2:**

500

**Solution:** Output 1. An off-the-shelf pre-trained model is not aligned to answer questions and will only generate likely continuations.

**Your answer:**

2. (3 points) Your colleagues have created two choices of in-context prompts for guiding your language model. Which of these prompts do you prefer and why?

| Prompt 1:                                  | Prompt 2:                                   |
|--------------------------------------------|---------------------------------------------|
| Input: Five hundred patients visited in... | Input: Five hundred patients visited in ... |
| Output: 500                                | Output: 500                                 |
| Input: The cost of two million...          | Paragraph: The cost of two million...       |
| Output: 2000000                            | Number: 2000000                             |
| Input: A thousand people gathered...       | Input: A thousand people gathered...        |
| Output: 1000                               | Result: 1000                                |

**Solution:** Prompt 1 since it uses consistent naming of the input/outputs.

**Your answer:**

3. (3 points) You give our colleague another chance! They came up with two in-context prompts to guide your language model. Which of these prompts do you prefer and why?

| Prompt 1:                                  | Prompt 2:                                  |
|--------------------------------------------|--------------------------------------------|
| Input: Five hundred patients visited in... | Input: Five hundred patients visited in... |
| Output: 500                                | Output: 500                                |
| Input: The cost of five hundred dollars... | Input: The cost of two million...          |
| Output: 500                                | Output: 2000000                            |
| Input: Five hundred people gathered...     | Input: A thousand people gathered...       |
| Output: 500                                | Output: 1000                               |

**Solution:** Prompt 2 since it uses a diverse set of outputs.

**Your answer:**

4. (4 points) Briefly explain how LoRA (Low-rank Adaptation) works.

**Solution:** LoRa (Low-Rank Adaptation) is a parameter-efficient tuning technique. LoRA adds an additive term to the original model's matrices:

$$W \leftarrow W + \Delta W,$$

The additive term is a low-rank matrix:

$$\Delta W = AB^T$$

For example, if  $\Delta W \in \mathbb{R}^{d \times d}$ ,  $A, B \in \mathbb{R}^{d \times h}$ , where  $h \ll d$ .

**Your answer:**

5. (3 points) Describe why Instruction-Tuning is done. (less than 3 sentences)

**Solution:** *There is a misalignment between the pre-training objective and what humans intend to see in the output of models. Instruction-tuning is a way to adapt models to human intentions by training models on input-output pairs.*

**Your answer:**

6. (4 points) Describe two limitations of Instruction-Tuning.

**Solution:**

- *Difficult to collect diverse data.*
- *Resulting models may not be good at open-ended generation tasks. Incentives word-by-word rote learning. That of their supervision data bounds the resulting LM's generality/creativity.*
- *Resulting models may hallucinate more regularly. Labeled data is collected agnostic to the LM's knowledge. Therefore, there might be a mismatch between labeled data and LM knowledge. Hence, we may be encouraging "hypocritic" behavior, which leads to further hallucinations*

**Your answer:**